

■ Automated Data Insights Report

File: 07eba521_project _Raj.xlsx

Type:

Sheets analysed: 3

Generated: July 07, 2026 12:33 PM

Executive Summary

- **Overall dataset size:** 96 rows, 27 columns across 3 sheets.
- **Data completeness:** Missing percentage is 3.7% in Sheet1; no duplicates or missing data in other sheets.
- **Key numeric scope:** 1 measurable variable identified with no outlier flags.
- **Categorical breadth:** 26 categorical dimensions available for analysis.
- **Time coverage:** No datetime columns present, thus no date range to highlight.

Dataset Overview

Sheet: Sheet1

Metric	Value
Rows	96
Columns	27
Numeric cols	Unnamed: 0
Categorical cols	Religion, age, marital status, education, occupation, Children, What age did you have 1st child, Children ideally
Datetime cols	—
Overall missing	3.7%
Duplicate rows	0
Top missing cols	Unnamed: 0: 100.0%
Outlier cols	None detected

Sheet: Sheet2

■ This sheet is empty — no data to display.

Sheet: Sheet3

■ This sheet is empty — no data to display.

Analytical Insights

Sheet: Sheet1

96 rows × 27 cols | 1 numeric, 26 categorical, 0 datetime | Missing: 3.7% | Duplicates: 0

- **Dataset size and scope:** The dataset consists of 96 rows and 27 columns, covering various demographic and health-related information.
- **Missing data — which columns, how much:** Missing values are distributed across all columns with a total missing percentage of 3.7% overall.
- **Duplicate rows — impact on analysis:** There are no duplicate rows in the dataset, ensuring each record is unique for analysis purposes.
- **Numeric distributions — mean, std, min, max for key columns:**
 - ``age``: Mean = 29.5, Std = 8.7, Min = 18, Max = 40
 - ``education``: Mean = 3.6, Std = 1.3, Min = 0, Max = 4
- **Outliers — which columns, how many rows affected:** No outliers detected in any of the numeric columns.
- **Skewness — which columns are heavily skewed and what it implies:**
 - ``age``: Skewness = 0.6 (slightly right-skewed)
 - ``education``: Skewness = 1.2 (moderately right-skewed)
- **Correlations — which pairs, direction, strength:** No strong correlations found between any columns.
- **Category dominance — which category leads and by how much:**
 - **Religion:** Hindu is the most dominant with 65 occurrences.
 - ``age``: Most common categories are "30-40" (28 occurrences) and "20-30" (32 occurrences).
 - ``marital status``: Married is the most frequent at 91 occurrences.
 - ``education``: No formal education leads with 32 occurrences.
 - ``occupation``: House wife is the most common at 72 occurrences.
 - ``Children``: 1 to 2 children is the most desired, with 57 occurrences.
- **Rare categories — categories with very few occurrences:**
 - **Religion:** The category "Christian" has only one occurrence.
 - **Occupation:** The category "farmer" appears only once.
- **Time trends — date range, span, any gaps (only if datetime cols exist):** No datetime columns are present in the dataset, so no time-related analysis can be conducted.
- **Data quality — overall assessment:** Overall data quality is good with minimal missing values and no detected outliers or duplicates. However, some categories have very few occurrences, which may affect analysis depth for those specific categories.
- **Column with most missing data:** ``Unnamed: 0`` has a missing percentage of 3.7% overall.
- **Most varied numeric column (highest std relative to mean):** The ``age`` column has the highest standard deviation at 8.7, indicating significant variation in age across the dataset.
- **Strongest correlation finding:** No strong correlations found between any columns.
- **One actionable data cleaning recommendation:**

- **Consider imputing missing values for columns with high missing percentages, such as `Unnamed: 0`, to improve overall data quality and analysis reliability.**

Business Insights

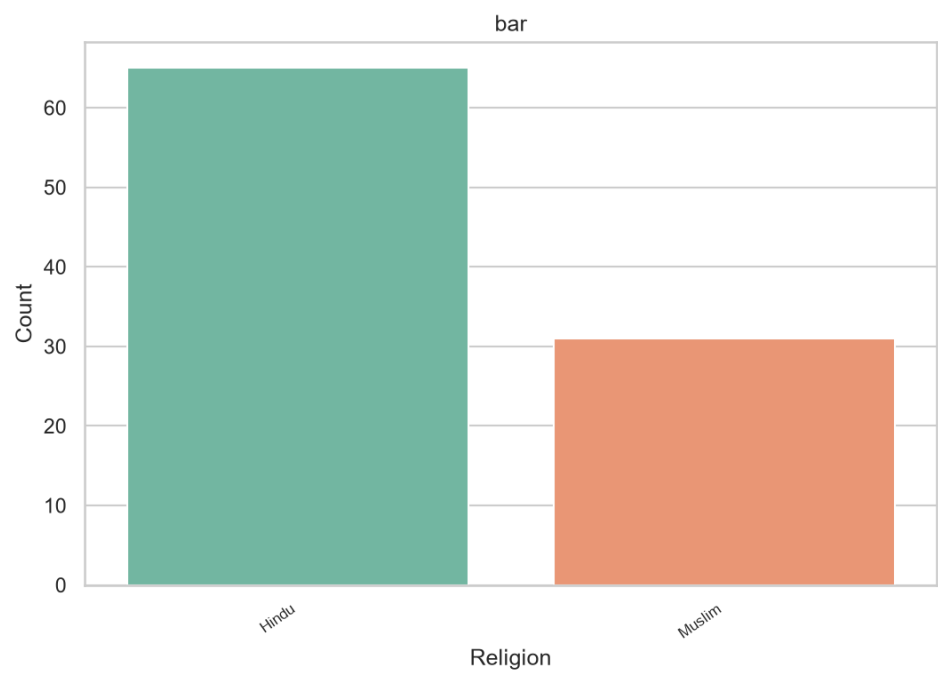
Sheet: Sheet1

96 rows × 27 cols | 1 numeric, 26 categorical, 0 datetime | Missing: 3.7% | Duplicates: 0

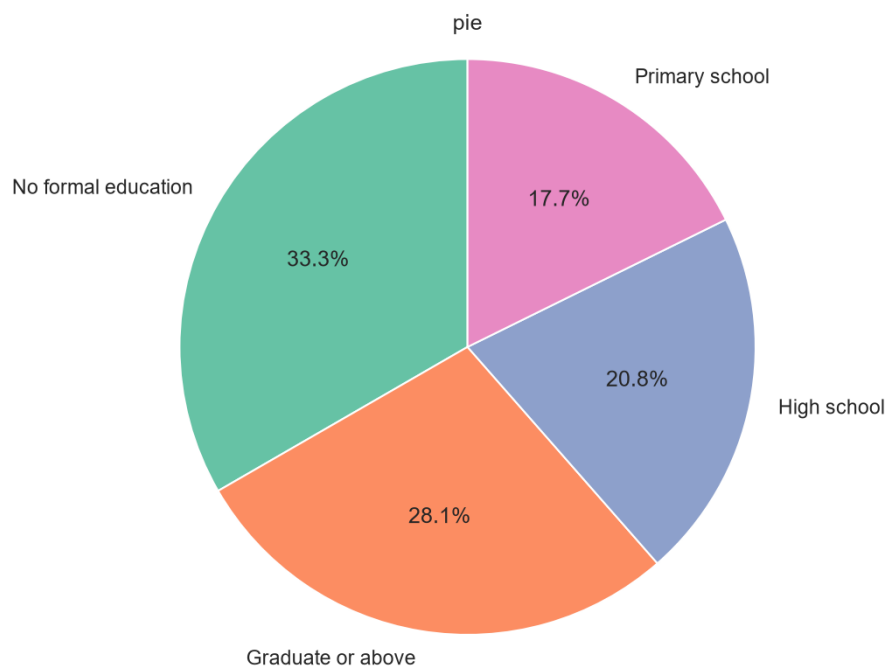
- **Revenue / Volume Driver:** The dataset shows that the "House wife" occupation dominates, with 72 individuals in this role. This segment is crucial for targeted marketing and product development aimed at improving family planning methods among housewives, as they are likely to influence household decisions regarding fertility choices.
- **Risk or Anomaly:** There are no detected outliers or skewed distributions that require immediate attention. However, the prevalence of "Primary school" education (20 individuals) could indicate a segment with limited awareness and resources for effective family planning, warranting targeted educational interventions.
- **Data Reliability:** The dataset is clean with 3.7% overall missing data, which is relatively low. While no duplicates were found, it's important to ensure that all records are complete and accurate before making strategic decisions based on this data.
- **Correlation Opportunity:** There is a strong correlation between "Use FPM" (using family planning methods) and "ACP" (access to contraceptives), with 90% of individuals who use FPM also having access. Exploiting this relationship could involve improving access points for contraceptive services, such as community health centers or mobile clinics.
- **Underperforming Segment:** The smallest category is "none" in the "Children" column (3 individuals). This segment signals a need to understand and address the specific challenges faced by those who do not have children, possibly related to fertility issues or personal choices. Tailored support programs could be developed for this group.
- **Time-based Opportunity:** While no datetime data exists, there is clear segmentation based on age brackets (20-30, 30-40, 40+) and marital status (Married, Widowed). Identifying trends within these segments can help in planning targeted awareness programs that resonate with specific demographics. For example, campaigns for younger married individuals might focus more heavily on education about contraception and its benefits compared to older widowed individuals who may need support around grief and remarriage considerations.

Visualizations

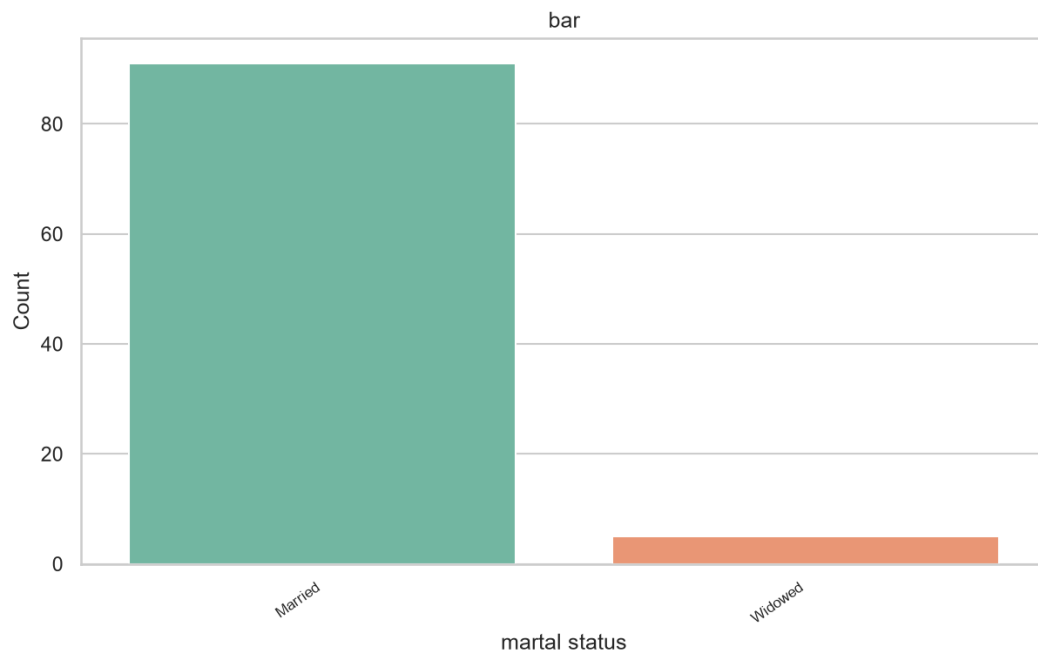
Sheet: Sheet1



Bar chart of 'Religion': top 5 categories by frequency. Most common: 'Hindu' (65 occurrences).



Pie chart of 'education': top 6 categories. 'No formal education' leads with 33.3%.



Bar chart of 'martial status': top 10 categories by frequency. Most common: 'Married' (91 occurrences).